

StreamLab Project Setup & Instructions

Welcome to the StreamLab project! This is a powerful Electron-Vite desktop application built with React, TypeScript, and TailwindCSS. It provides a visual block-based programming interface (powered by Blockly) alongside a dynamic Hardware Canvas to seamlessly interact with microcontrollers like Arduino Uno, Arduino Mega, and ESP32.

Prerequisites

Before running the project from the source code, please ensure you have the following installed on your system:

1. **Node.js** (v18 or higher recommended) - Download from <https://nodejs.org/>
2. **Python & C++ Build Tools** - Since this project uses the `serialport` package which relies on native C++ bindings, you must have Windows Build Tools installed. If you don't have them, you can install them via the Node installer or by running `npm install --global windows-build-tools` in an administrative terminal.

How to Run the App Locally

1. **Extract the ZIP file** to a folder of your choice.
2. Open a terminal (Command Prompt or PowerShell) and navigate into the extracted project folder:

```
cd streamlab-project
```

3. **Install the dependencies:**

```
npm install
```

4. **Start the development server:**

```
npm run dev
```

This will immediately launch the Electron app in development mode with Hot-Module Replacement (HMR) enabled.

How to Build the Executable (.exe)

If you want to package the application into a standalone installer for Windows so others can install it without needing Node.js:

1. Open a terminal in the project directory.
2. Run the packaging command:

```
npm run package
```

3. Once the build process successfully completes, you will find the generated `.exe` installer file inside the `dist` folder:

```
streamlab-project/dist/my-stream-lab Setup 1.0.0.exe
```

Project Structure Overview

- `src/main/` - The Electron main process code. This handles the heavy lifting: Arduino CLI execution, toolchain installation, serial port communication, and window management.
- `src/renderer/` - The React frontend code. This handles all the UI, including the Interactive Hardware Canvas, the Blockly workspace, and the code editor.
- `src/shared/` - Shared types, board configurations, and wiring logic that are used by both the main and renderer processes.
- `tests/` - Comprehensive unit tests for the hardware generator logic.

Troubleshooting

- **serialport installation errors:** If `npm install` fails on the `serialport` package or `@serialport/bindings-cpp`, ensure your Visual Studio C++ Build Tools and Python are correctly installed and added to your system PATH.
- **Missing Arduino CLI / Toolchain:** The app is designed to automatically download the Arduino CLI and required cores to your local AppData directory (`AppData/Roaming/my-stream-lab/Arduino15`). Ensure your internet connection is active on the first run and that your firewall is not blocking the downloads.
- **Blank Screen on Startup:** Check your terminal where you ran `npm run dev` for any compilation errors.

Enjoy building and experimenting with StreamLab!